Logistics Document Intelligence RAG

Course Name: Generative AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Academic Year:2026

Problem Statement & Objectives

1. Problem Statement:

Chat with Logistics Documents (RAG)

Description of problem Statement:

Architect a complete Retrieval-Augmented Generation (RAG) system for Logistics document intelligence. The solution enables users to upload professional documents (PDF reports, manuals) which are then processed through a pipeline of text chunking and vector embedding. By leveraging a high-performance Vector Database and LLM APIs, the system facilitates accurate, context-aware Q&A sessions, ensuring that responses are grounded strictly in the provided domain-specific content.

Logistics is full of a wide variety of unstructured documents, such as invoices, shipping documents, delivery receipts, customs declarations, bills of lading, and many others. These types of documents contain critical operational information, such as invoice numbers, shipping dates, delivery addresses, and payment methods, which are all very useful to companies in the logistics industry. Unfortunately, most of the presently used capabilities for retrieving information contained within these documents rely on manual effort through processes such as reading through entire files to locate information, or performing very basic keyword searches.

Analyzing these documents manually is very labour-intensive and time-consuming; moreover, due to the volume of documents being analysed, there are numerous opportunities for human error. Additionally, current keyword-based search systems do not have the ability to understand the semantic meanings of a user's query, and therefore, may not be able to provide users with the information they are looking for in an unstructured document, even though that information exists.

Also, although large language models (LLMs) can answer natural language questions based on their significant amount of knowledge, when they are not connected to the actual data contained within the documents, they could generate incorrect or "hallucinated" responses.

It is apparent that a successful intelligent document processing system must:

- Interpret natural language questions.
- Identify the relevant data within logistics documents.
- Provide a response that is accurate and contextually relevant.
- Reduce the amount of manual effort expended and increase productivity.

This Logistics Document Intelligence System will solve the above problem by combining Semantic Search (using a Vector Database) and Answer Generation (using a Large Language Model) through Retrieval-Augmented Generation (RAG) to produce trusted and accurate answers from logistics documents.

2. Project Objectives

- **To develop a document processing pipeline for logistics PDFs**
  To design and implement a system that allows users to upload logistics documents and automatically extract and divide the text into smaller chunks for efficient processing and retrieval.

- **To implement vector embedding and storage using a vector database**
  To generate semantic vector embeddings of document text using the SentenceTransformer model and store them in a FAISS vector database to enable fast and accurate semantic similarity search.

- **To implement semantic retrieval based on user queries**
  To convert user queries into vector embeddings and retrieve the most relevant document chunks from the vector database using similarity matching techniques.

- **To integrate a Large Language Model for context-aware answer generation**
  To integrate Large Language Model API to generate accurate answers using the retrieved document context, ensuring responses are grounded in the uploaded logistics documents.

3. Scope of the Project

The objective of this project is to create and deploy a document intelligence system using retrieval-augmented generation (RAG) technology that will allow intelligent answers to be given to questions posed about uploaded logistics documents.

The complete system will allow users to input and retrieve data from logistics documents using vector embeddings and large language model APIs. In addition, the complete document intelligence pipeline will include the ability to process data from logistics documents, store them as reference points, retrieve them when needed, and generate response(s) based on the information contained in them.

These are the main areas of focus for this project:

- **Uploading and processing documents**
  The system allows users to upload documents related to logistics that are stored in PDF format. Example documents are shipment reports, invoices, and manuals. Once an uploaded document is processed, the system will extract the text from the document. After that the text will be broken into smaller pieces so that semantic analysis can be performed more efficiently.

- **Embedding and storing text as vectors**
  The project creates semantic vector embeddings from the text of the documents that are uploaded using the SentenceTransformer model. Embeddings will be stored in a FAISS vector database so that documents can be retrieved quickly and efficiently based on similarity.

- **Query processing and retrieval using semantics**
  The system will accept natural language queries from users. Each query will be transformed into vector form and compared with the embeddings of the uploaded documents to return structured document chunks as the most similar to the query based on semantics.

- **Answering questions using a large language model**
  The project will also provide integration with the Large Language Model API to generate accurate and contextually relevant answers from the retrieved document content that are based on or "grounded" in the uploaded logistics documents

- **User interface for asking questions**
  The system's user interface provides users with the ability to upload documents, ask questions, and receive answers from the system as well as confidence scores for the answers and citations to the documents from which the answers came.

Project Scope Limitations

The projects have few limitations as follows:

- Only PDF document formats are supported within the system
- The vector database has used FAISS and is stored on the computer
- The system's intent is for academic/prototyping purposes only
- The system relies on an external LLM API (Groq) and access to the internet
- The system was only tested with logistics documents

In this project, we have covered the implementation of the RAG-based Logistics Document Question and Answering System. This includes document processing, vector storage and retrieval; semantic retrieval and grounded (or fact-based) answers through the use of a large language model.

Proposed Solution

A Logistics Document Intelligence System, based upon Retrieval-Augmented Generation (RAG), is proposed as a means for users to upload images of logistics-related documents and ask intelligent questions using advanced semantic search utilizing a Large Language Model (LLM).

The proposed solution receives user-uploaded PDF files, and processes the uploaded PDF file by converting it to vector embeddings; these vector embeddings are stored in a vector database, and when the user asks a question, the LLM is called to retrieve the relevant vector embedding(s) so that content from the document can be sent to the Groq LLaMA-3 LLM to provide an intelligent, contextualized, accurate response, based on what is contained within the document.

The proposal is intended to facilitate an efficient, accurate and trustworthy means of extracting information from logistics-related documents while minimising manual effort required to do so.

1. Key Features

- Upload logistics documents in PDF format
- Automatic text extraction and chunking
- Semantic vector embedding generation
- Vector storage using FAISS vector database
- Semantic similarity search based on user query
- Context-aware answer generation using Groq LLaMA-3
- Confidence score calculation for answers
- Display of document source citation
- Interactive user interface using Streamlit

2. Overall Architecture / Workflow

The High-Level Architecture Diagram shows the overall structure and interaction between different components of the Logistics Document Intelligence System.

The user interacts with the system through the Streamlit frontend, which provides the interface for uploading documents and asking questions. The frontend communicates with the Python backend, which controls the entire system.

The backend processes documents, generates embeddings using the SentenceTransformer model, and stores them in the FAISS vector database for semantic retrieval.

When a user submits a query, the backend retrieves relevant context from FAISS and sends it to the Groq LLaMA-3 model. The model generates an accurate answer based on the retrieved document content, which is then displayed to the user.

This architecture ensures efficient document processing, accurate semantic search, and reliable answer generation using the Retrieval-Augmented Generation (RAG) approach.

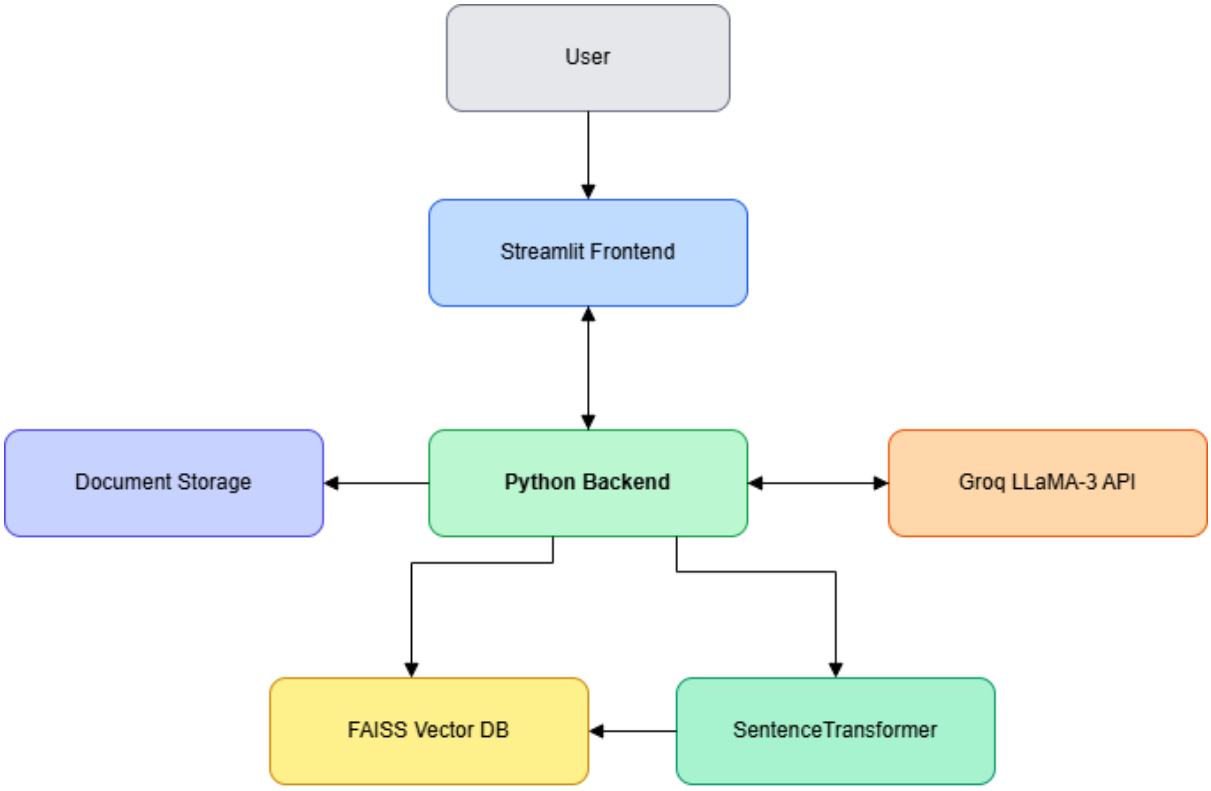

Figure 1: High-Level Architecture of the System

Workflows

(i). Document Processing Workflow

The document processing workflow starts when the user uploads a logistics PDF through the Streamlit frontend. The Python backend extracts the text from the document and divides it into smaller chunks for efficient processing.

These chunks are converted into vector embeddings using the SentenceTransformer model. The embeddings are then stored in the FAISS vector database, which allows the system to perform fast semantic search.

This workflow prepares the document data for accurate retrieval and question answering.

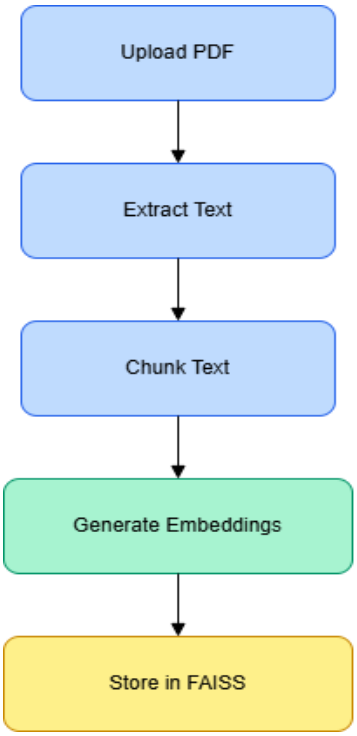

Figure 2: Document Processing Workflow

(ii). Query Processing Workflow

The query processing workflow begins when the user enters a question through the Streamlit interface. The Python backend converts the query into a vector embedding using the SentenceTransformer model.

The system searches the FAISS vector database to retrieve the most relevant document chunks based on semantic similarity. This retrieved context is sent to the Groq LLaMA-3 model, which generates an accurate answer.

The final answer, along with confidence score and citation, is displayed to the user through the Streamlit frontend.

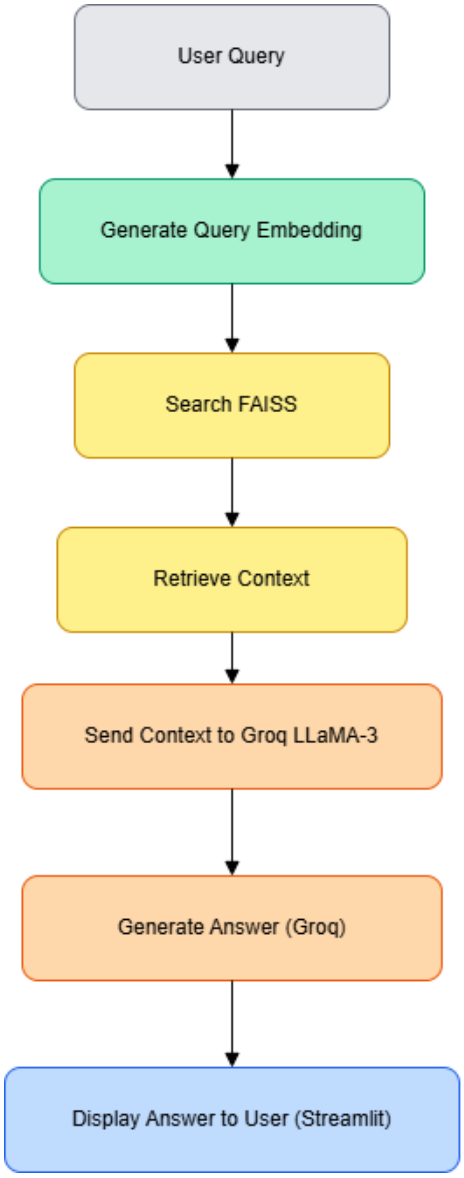

Figure 3: Query Processing Workflow

3. Tools & Technologies Used

The following tools and technologies were used to develop the Logistics Document Intelligence System using Retrieval-Augmented Generation:

Table 1:Tools and Technologies Used in the Development of the RAG-based Logistics Document Intelligence System

Tool / Technology	Category	Purpose
Python	Programming Language	Used to develop the backend logic and implement the RAG pipeline
Streamlit	Frontend Framework	Used to create the user interface for document upload and question answering
SentenceTransformer (all-MiniLM-L6-v2)	Embedding Model	Used to convert document text and queries into vector embeddings
FAISS (Facebook AI Similarity Search)	Vector Database	Used to store embeddings and perform fast semantic similarity search
Groq LLaMA-3 API	Large Language Model	Used to generate accurate answers based on retrieved document context
PyPDF / PDF Loader	Document Processing	Used to extract text from uploaded PDF documents
VS Code	Development Environment	Used for coding, debugging, and running the application
Local File System	Storage	Used to store uploaded PDF files and FAISS index locally

Results & Output

1. Screenshots / Outputs

The following section provides interface screenshots and model-generated responses that validate the successful implementation of the Retrieval-Augmented Generation (RAG)–based Logistics Document Intelligence System.

(i). System Home Interface

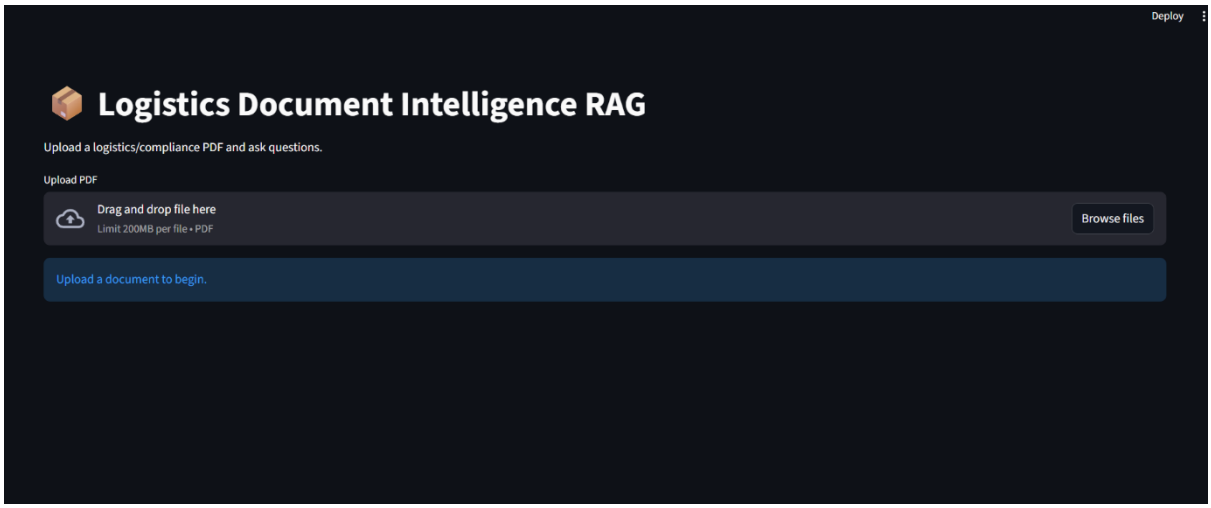

Figure 4: Initial User Interface of the Logistics Document Intelligence System

This screenshot shows the main interface of the system developed using Streamlit. It provides the option to upload logistics PDF documents and interact with the system. This interface serves as the entry point for users to begin document processing and question answering.

(ii). Document Upload and Indexing

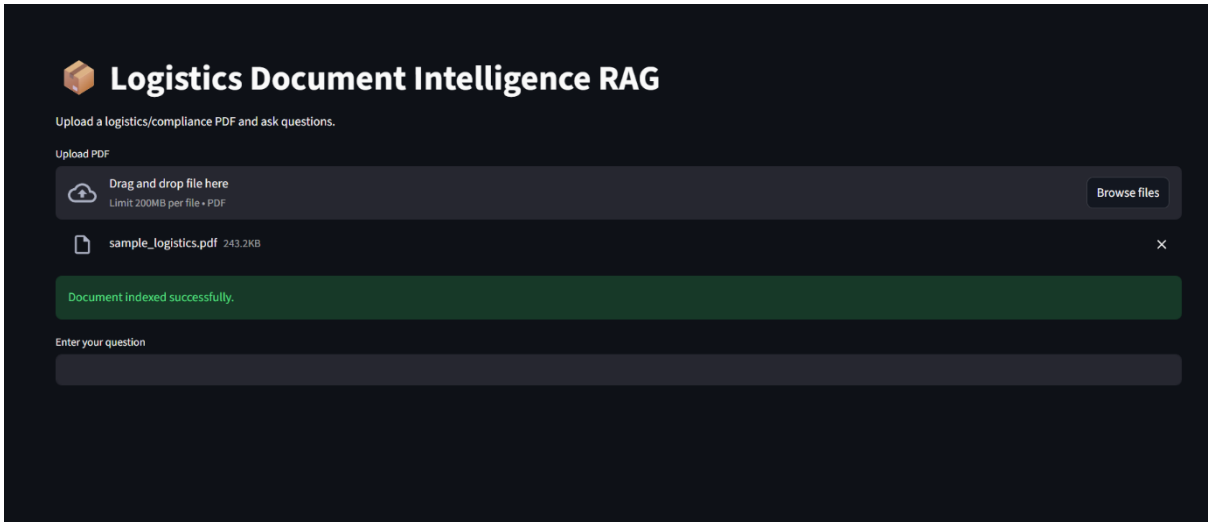

Figure 5: Successful Upload and Indexing of Logistics PDF Document

This screenshot shows that the logistics document (sample_logistics.pdf) was successfully uploaded and indexed. The message "Document indexed successfully" confirms that the system extracted text, generated embeddings using the SentenceTransformer model, and stored them in the FAISS vector database.

This step prepares the document for semantic search and retrieval.

(iii). Query Input and Answer Generation

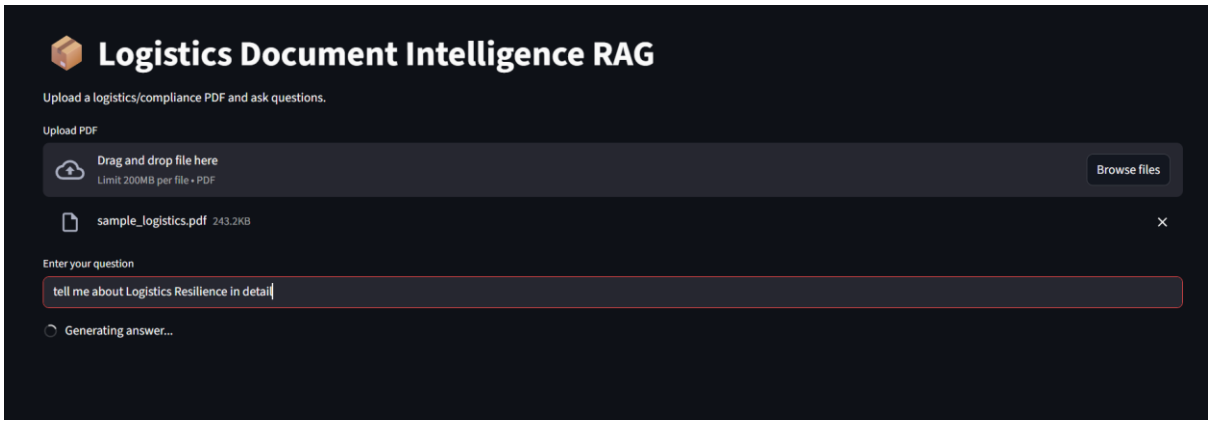

Figure 6: User Query and Generated Answer using Groq LLaMA-3

This screenshot shows the user entering a natural language question related to Logistics Resilience. The system processed the query and generated a context-aware answer using the Groq LLaMA-3 Large Language Model based on the uploaded document.

This demonstrates the successful implementation of Retrieval-Augmented Generation.

(iv). Answer and Confidence Score Display

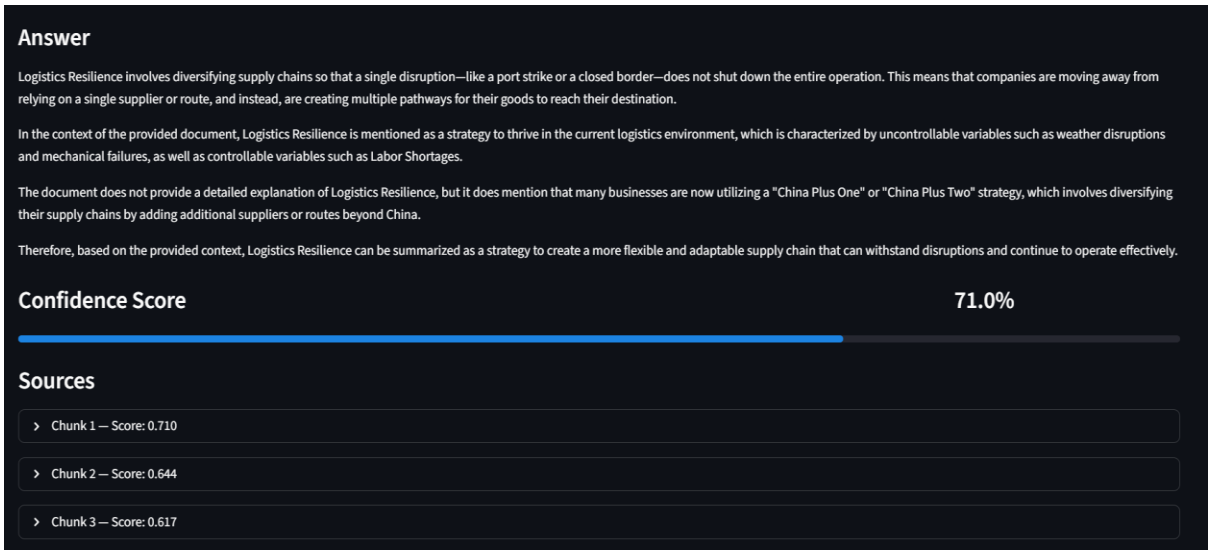

Figure 7: Answer along with Confidence Score of Generated Answer

This screenshot shows the confidence score of 71.0%, which indicates the reliability of the generated answer. The confidence score is calculated based on semantic similarity between the user query and retrieved document chunks from the FAISS vector database.

This helps users evaluate the accuracy of the system's response.

(v). Source Citation and Retrieved Document Chunks

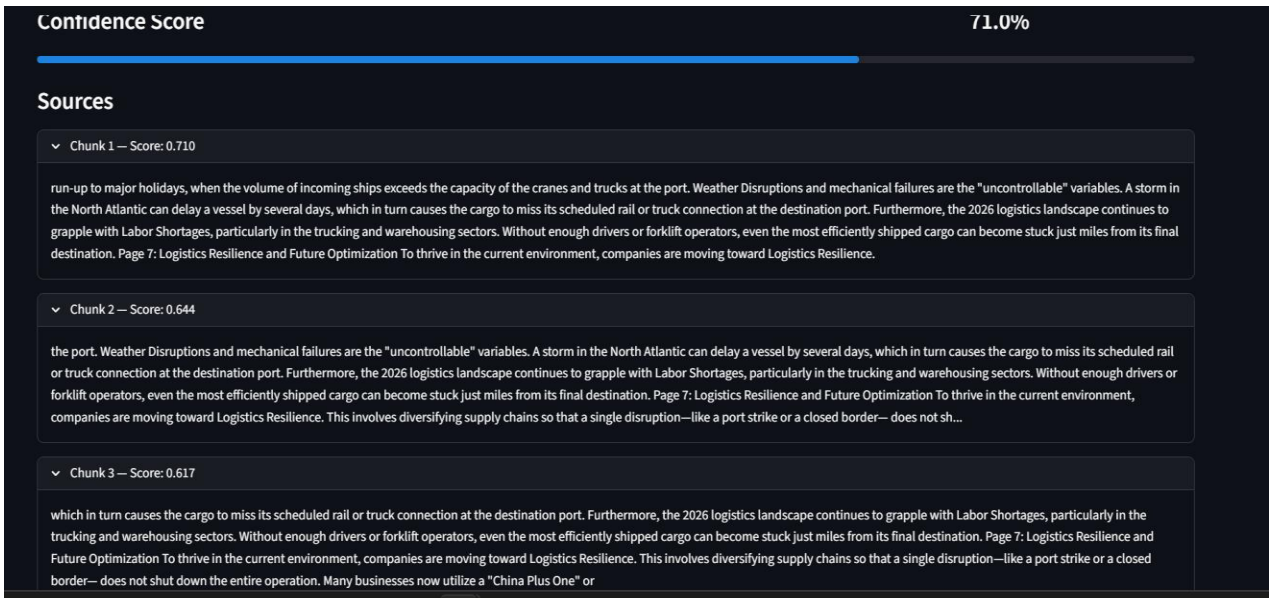

Figure 8: Retrieved Document Sources and Similarity Scores

This screenshot shows the document chunks retrieved from the FAISS vector database along with their similarity scores. These chunks are used as context by the Groq LLaMA-3 model to generate the answer.

Displaying sources improves transparency and ensures that the answer is grounded in the original document.

The system successfully performs document upload, embedding generation, semantic search, and answer generation using Retrieval-Augmented Generation. The system also provides confidence scores and document citations, ensuring reliable and transparent results.

2. Reports / Dashboards / Models

The Logistics Document Intelligence System includes an interactive dashboard and integrated AI models to enable intelligent document retrieval and question answering.

Streamlit Dashboard

The system uses a Streamlit-based dashboard that allows users to upload logistics documents, enter queries, and view generated answers in real time. The dashboard displays the generated

answer along with a confidence score and source citations. This provides a complete and user-friendly interface for interacting with the system.

The dashboard also shows document indexing status, confirming that the document has been successfully processed and stored in the vector database.

Embedding Model

The system uses the SentenceTransformer (all-MiniLM-L6-v2) model to convert document text and user queries into vector embeddings. These embeddings capture the semantic meaning of the text and enable accurate similarity comparison during retrieval.

Vector Database Report (FAISS)

The FAISS vector database stores the generated embeddings and performs semantic similarity search. It retrieves the most relevant document chunks based on the user query and provides similarity scores, which are used to calculate the confidence score.

The retrieved document chunks are also displayed in the dashboard as source references.

Large Language Model (Groq LLaMA-3)

The Groq LLaMA-3-8B Large Language Model generates accurate and context-aware answers using the retrieved document context. This ensures that the generated response is relevant to the uploaded document.

System Output Report

The system produces the following outputs:

- Generated answer to user query
- Confidence score indicating answer reliability
- Source document chunks with similarity scores

These outputs confirm that the system successfully performs intelligent document retrieval and question answering using Retrieval-Augmented Generation.

3. Key Outcomes

The implementation of the Logistics Document Intelligence System using Retrieval-Augmented Generation (RAG) successfully achieved the following outcomes:

- Successful Development of RAG-based Document Intelligence System
- Accurate Semantic Retrieval of Document Content
- Context-Aware Answer Generation using Groq LLaMA-3
- Confidence Score and Source Citation for Reliability
- User-Friendly Interactive Dashboard
- Reduction of Manual Effort in Document Analysis and increases efficiency

The project successfully demonstrated the practical implementation of a Retrieval-Augmented Generation system for logistics document intelligence, capable of providing accurate, reliable, and context-aware answers from uploaded documents.

Conclusion

The designers successfully looked to build a Logistics Document Intelligence System that would allow users to intelligently ask questions on logistics based PDF documents. The system allows users to upload the documents, performs text extraction and embedding generation, and saves the embeddings in a FAISS vector database for efficient semantic searching.

When a user submits a question, it retrieves relevant documents and uses the Groq LLaMA-3 Large Language Model to generate an accurate & contextually relevant answer. It also provides confidence scores and citations to allow easy reference and validate the responses produced. The system's user/validator interface was created with Streamlit, therefore demonstrating the complete end-to-end process.

Through this project, we have gained a practical understanding of Retrieval-Augmented Generation; I have gained hands on experience building real world applications involving this technology and the integration of vector databases, embedding models, and large language models.

The project has shown a feasible and dependable logistical intelligence system to provide a user with an efficient and dependable way of retrieving information from logistics documents while at the same time reducing the amount of manual labour required to read through documents to retrieve information using modern day artificial intelligence techniques.

Future Scope & Enhancements

The current Logistics Document Intelligence System provides accurate document retrieval and question answering using Retrieval-Augmented Generation. However, the system can be further enhanced with the following improvements:

1. Support for Multiple Document Formats

The system can be extended to support additional file formats such as Word documents (DOCX), Excel files (XLSX), and text files (TXT). This will make the system more flexible and useful for different types of business documents.

2. Implementation of Advanced Vector Databases

The current system uses FAISS for local storage. In the future, it can be integrated with enterprise-level vector databases such as Pinecone, Weaviate, or Chroma, which provide better scalability, cloud storage, and faster retrieval for large datasets.

3. Deployment as a Web Application

The system can be deployed on cloud platforms such as AWS, Azure, or Google Cloud to make it accessible to users from anywhere. This will allow real-world organizations to use the system in production environments.

4. Integration with More Advanced Language Models

More advanced Large Language Models can be integrated to further improve answer accuracy, reasoning capability, and response quality.

5. Multi-Document and Large Dataset Support

The system can be enhanced to handle large volumes of documents and support querying across multiple documents simultaneously.

6. User Authentication and Security

Security features such as user login, authentication, and access control can be added to protect sensitive business documents.

These enhancements will improve the scalability, usability, and performance of the system, making it suitable for real-world enterprise applications in logistics and document intelligence.